

Wiring:

ESP32-S3 Tenstar Pin	ST7796S / ILI9488 Display Pin	Touch Pin	SD Card Pin	Buzzer
5V / VBUS	VCC	VCC if required	VCC if required	-
3.3V	LED / BL	VCC if touch is 3.3V only	VCC if SD module is 3.3V only	-
GND	GND	GND	GND	Buzzer -
GPIO8 (A5)	CS	-	-	-
GPIO14 (A4)	RESET / RST	-	-	-
GPIO15 (A3)	DC / RS	-	-	-
GPIO11 (D11 / MOSI)	SDI / MOSI	T_DIN	-	-
GPIO12 (D12 / SCK)	SCK / CLK	T_CLK	-	-
GPIO13 (D13 / MISO)	SDO / MISO if used	T_DO	-	-
GPIO10 (D10)	-	T_CS	-	-
GPIO6 (D6)	-	T_IRQ	-	-
GPIO9 (D9)	-	-	SD_CS	-
GPIO5(D5)	-	-	-	Buzzer +
GPIO35(MOSI)	-	-	SD_MOSI	-
GPIO36(SCK)	-	-	SD_SCK	-
GPIO37(MISO)	-	-	SD_MISO	-

ESP32-S3 Tenstar Pin	LC29H Pin
GPIO17 (A1 / ESP32 RX)	Pin 8 Pi TXD0 / LC29H TX
GPIO18 (A0 / ESP32 TX)	Pin 10 Pi RXD0 / LC29H RX
GND	Pin 6 GND
Not connected	LC29H VCC if LC29H is powered by USB
Power bank USB output	LC29H USB power

Note: The RX and TX pins may need to be swapped on LC29H, some boards mislabel them

Internal ESP32-S3 GPIO	Internal Device Pin
GPIO7	Onboard ST7789 CS
GPIO39	Onboard ST7789 DC
GPIO40	Onboard ST7789 RESET
GPIO45	Onboard ST7789 Backlight
GPIO35	Onboard ST7789 MOSI
GPIO36	Onboard ST7789 SCK
GPIO37	Onboard ST7789 MISO

SPI Function	ESP32-S3 Tenstar Pin	Shared By
MOSI	GPIO11 (D11 / MOSI)	Display, Touch, SD Card
SCK	GPIO12 (D12 / SCK)	Display, Touch, SD Card
MISO	GPIO13 (D13 / MISO)	Touch, SD Card
Display CS	GPIO8 (A5)	Display only
Touch CS	GPIO10 (D10)	Touch only
SD CS	GPIO9 (D9)	SD Card only